

Table 1. RealToxicityPrompts results with evaluated-model KV cache disabled. Values are mean $\pm$ standard error across five repetitions; TBD marks unrun model-method pairs.

	Method	Toxic continuations (%) $\downarrow$	Distinct-2 $\uparrow$	Perplexity $\downarrow$	Safe-concept relevance (0-2) $\uparrow$	Instruction relevance (0-2) $\uparrow$	Fluency (0-2) $\uparrow$	Overall steering (0-2) $\uparrow$
Gemma-2-2B	Original	3.80 ± 0.26	0.657 ± 0.000	10.93 ± 0.05	0.72 ± 0.02	1.42 ± 0.01	1.42 ± 0.00	0.68 ± 0.02
	ITI	TBD	TBD	TBD	TBD	TBD	TBD	TBD
	ActAdd	TBD	TBD	TBD	TBD	TBD	TBD	TBD
	Mean-AcT	TBD	TBD	TBD	TBD	TBD	TBD	TBD
	Linear-AcT	TBD	TBD	TBD	TBD	TBD	TBD	TBD
	PID-AcT	TBD	TBD	TBD	TBD	TBD	TBD	TBD
	ODESteer	TBD	TBD	TBD	TBD	TBD	TBD	TBD
	S-PID	0.26 ± 0.08	0.684 ± 0.001	11.12 ± 0.06	0.56 ± 0.02	<u>1.48 ± 0.01</u>	<u>1.44 ± 0.01</u>	0.53 ± 0.02
	A-LQR	0.00 ± 0.00	0.629 ± 0.001	24.44 ± 0.20	0.29 ± 0.04	1.27 ± 0.00	1.09 ± 0.00	0.24 ± 0.03
	H_∞ (ours)	0.88 ± 0.09	0.694 ± 0.001	10.99 ± 0.02	0.72 ± 0.05	1.46 ± 0.01	1.40 ± 0.01	0.67 ± 0.05
Llama-3-8B	Original	4.92 ± 0.71	0.676 ± 0.001	7.66 ± 0.03	0.75 ± 0.02	1.45 ± 0.01	1.58 ± 0.01	0.70 ± 0.03
	ITI	TBD	TBD	TBD	TBD	TBD	TBD	TBD
	ActAdd	TBD	TBD	TBD	TBD	TBD	TBD	TBD
	Mean-AcT	TBD	TBD	TBD	TBD	TBD	TBD	TBD
	Linear-AcT	TBD	TBD	TBD	TBD	TBD	TBD	TBD
	PID-AcT	TBD	TBD	TBD	TBD	TBD	TBD	TBD
	ODESteer	TBD	TBD	TBD	TBD	TBD	TBD	TBD
	S-PID	0.84 ± 0.09	<u>0.685 ± 0.001</u>	<u>7.85 ± 0.04</u>	<u>0.66 ± 0.04</u>	<u>1.50 ± 0.01</u>	1.62 ± 0.01	<u>0.65 ± 0.04</u>
	A-LQR	<u>0.14 ± 0.07</u>	0.682 ± 0.001	8.50 ± 0.03	0.63 ± 0.03	1.48 ± 0.02	<u>1.63 ± 0.01</u>	0.59 ± 0.03
	H_∞ (ours)	0.14 ± 0.05	0.648 ± 0.002	11.01 ± 0.05	0.44 ± 0.03	1.48 ± 0.02	1.25 ± 0.01	0.40 ± 0.02
Qwen-2.5-72B	Original	TBD	TBD	TBD	TBD	TBD	TBD	TBD
	ITI	TBD	TBD	TBD	TBD	TBD	TBD	TBD
	ActAdd	TBD	TBD	TBD	TBD	TBD	TBD	TBD
	Mean-AcT	TBD	TBD	TBD	TBD	TBD	TBD	TBD
	Linear-AcT	TBD	TBD	TBD	TBD	TBD	TBD	TBD
	PID-AcT	TBD	TBD	TBD	TBD	TBD	TBD	TBD
	ODESteer	TBD	TBD	TBD	TBD	TBD	TBD	TBD
	S-PID	TBD	TBD	TBD	TBD	TBD	TBD	TBD
	A-LQR	TBD	TBD	TBD	TBD	TBD	TBD	TBD
	H_∞ (ours)	TBD	TBD	TBD	TBD	TBD	TBD	TBD